APEX GLOBAL BANK

SR 11-7 MODEL DEVELOPMENT DOCUMENT

Market Risk Value-at-Risk Model

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Owner	Dr. Priya Nair (CRO)
Validator	Dr. Samuel Achebe (Model Validation Officer)
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Regulatory Framework: 12 CFR 3.132, Basel III Annex 10, SR 11-7

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1 Model Overview

1.1 Purpose

The Market Risk VaR Model quantifies the potential loss in the trading book over a 1-day horizon at 99% confidence. It is the primary input to regulatory market risk capital under 12 CFR 3.132 (Basel III IMA) and drives desk-level risk limits in LimitManager.

1.2 Business Use

- **Regulatory capital:** Basel multiplier 3× on the higher of current VaR and 60-day average VaR.
- **Risk limits:** Daily delta and vega utilisation reported against board-approved limits.
- **Pre-trade checks:** Parametric VaR delta used for pre-trade limit approval in OMS.
- **Stress testing:** Inputs to DFAST adverse scenario projections.

1.3 Scope

All instruments in the Apex trading book: AAPL, MSFT, SPY, NVDA (equities), EURUSD, GBPUSD (FX), US10Y, US2Y (rates), AAPL_CALL_200 (equity option), USD_IRS_5Y (interest rate swap).

2 Theoretical Basis

2.1 GBM Assumption

Asset returns are modelled as log-normal under geometric Brownian motion:

$$dS_t = \mu S_t \, dt + \sigma S_t \, dW_t$$

For a 1-day horizon, the log-return is approximately normal: $r_{1d} \sim \mathcal{N}(0, \sigma^2)$.

2.2 Three VaR Methodologies

2.2.1 Method 1 — Historical Simulation (Primary)

$$\mathrm{VaR}_{\mathrm{HS}} = -\mathrm{Percentile}(P\&L_t,\ 1\%), \quad t \in \{t-1, \ldots, t-250\}$$

Full revaluation of current portfolio under 250 historical 1-day scenarios. No distributional assumption.

2.2.2 Method 2 — Parametric Delta-Normal

$$\mathrm{VaR}_{\mathrm{param}} = 1.645 \, \sigma_{\mathrm{portfolio}} \sqrt{W}$$

where $\sigma_{\mathrm{portfolio}} = \sqrt{\Delta^\top \Sigma \, \Delta}$, Δ is the vector of delta sensitivities, Σ is the return covariance matrix, and W is the notional weight.

2.2.3 Method 3 — Monte Carlo Cholesky (Regime-Aware)

$$
\begin{aligned}
L &= \mathrm{Chol}(\Sigma_{\text{regime}}) \\
Z_t &\sim \mathcal{N}(0, I) \\
r_t &= \mu + L Z_t \\
\text{P\&L} &= \sum_i \Delta_i \cdot r_{t,i} \cdot S_i \\
\mathrm{VaR}_{\mathrm{MC}} &= -\mathrm{Percentile}(\text{P\&L},\ 1\%) \quad \text{over } 2\,000 \text{ paths}
\end{aligned}
$$

The correlation matrix Σ_{regime} is selected by the Correlation Regime Model (APEX-MDL-0009): NORMAL or STRESS depending on realized equity-credit correlation.

2.3 Basel Multiplier

$$
\text{Capital} = \max\bigl(\mathrm{VaR}_t,\ k\ \overline{\mathrm{VaR}}_{60}\bigr) \times \sqrt{10}, \qquad k = 3.0 + \text{add-on}
$$

The add-on ranges from 0 to 1.0 based on backtesting exceptions (BCBS traffic-light).

3 Mathematical Specification

Parameter	Value	Source
Confidence level	99%	12 CFR 3.132
Horizon	1-day (scaled to 10-day for capital)	Basel III
Historical window	250 business days	SR 11-7
MC paths	2 000	Internal calibration
Basel multiplier k	3.0 (base)	12 CFR 3.132(d)
10-day scaling	$\times\sqrt{10}$	Basel III

3.1 Backtesting Traffic-Light (BCBS)

Exceptions in 250 days	Zone	Add-on
0–4	Green	0.00
5–9	Amber	0.40–0.85
≥ 10	Red	1.00

4 Implementation

Code location: `infrastructure/risk/var_calculator.py`
Class: `VaRCalculator`

Key methods:

- `calculate(positions, prices)` — Monte Carlo VaR, regime-aware.
- `calculate_parametric(positions, prices)` — delta-normal for pre-trade.

Calibration: Covariance matrix estimated from 60-day exponentially-weighted returns ($\lambda = 0.94$). Regime selection from `CorrelationRegimeModel.detect_regime()`.

Runtime: ∼50 ms for 2 000-path MC on 11 instruments.

5 Validation

Backtesting methodology: Compare daily P&L realisation against prior-day VaR. Exceptions recorded in exception log. Traffic-light assessment quarterly.

Independent replication: Validator replicated MC VaR using NumPy; confirmed within ±2% for equity-only portfolio. IRS DV01 sensitivity confirmed against Bloomberg SWPM.

Benchmarking: Parametric VaR benchmarked against RiskMetrics delta-normal; within ±5% for equity portfolio.

6 Model Limitations

1. **Normality assumption (parametric):** Fat tails not captured; parametric VaR underestimates tail risk in stressed markets by 20–40% (empirical observation, GFC window).
2. **250-day window sensitivity:** In low-volatility regimes, VaR may understate risk if the window excludes stress events. Compensating control: SVaR (APEX-MDL-0002) adds stressed capital buffer.
3. **Square-root-of-time scaling:** Assumes IID returns; serial correlation violates this assumption for fixed income instruments.
4. **2 000-path MC estimation error:** Sampling noise at the 99th percentile ≈ ±8% at 2 000 paths. Partially mitigated by antithetic variates.

7 Open Findings

ID	Severity	Description	Status
VAR-F1	**Major**	Look-back window sensitivity not stress-tested across multiple window lengths (125d, 250d, 500d).	Open
VAR-F2	**Minor**	Parametric method uses flat normal assumption; Cornish-Fisher expansion not applied.	Open